Forensic Science International: Digital Investigation 32 (2020) 200899

Contents lists available at ScienceDirect

Forensic Science International: Digital Investigation

journal homepage: www.elsevier.com/locate/fsidi

Digital image forensic approach based on the second-order statistical analysis of CFA artifacts

Gurinder Singh, Kulbir Singh*

Department of Electronics and Communication Engineering, TIET, Patiala, Punjab, India

ARTICLE INFO

Article history:
Received 17 June 2019
Received in revised form
20 November 2019
Accepted 20 December 2019
Available online xxx

Keywords:
Re-interpolation process
Demosaicing
Markov transition probability matrix
Machine learning

ABSTRACT

The truthfulness of digital images can be evaluated by investigating the CFA artifacts introduced due to the interpolation process of the image acquisition phase. In this paper, an image tampering detection technique is proposed by exposing the CFA artifacts in difference domain through the higher-order statistical analysis based on the Markov transition probability matrix (MTPM). Firstly, the given image is re-interpolated with most commonly used four Bayer CFA patterns. The re-interpolation process is performed by using bilinear interpolation scheme for simplicity purpose. Then, the difference between the given image and its re-interpolated versions is evaluated to analyze the CFA inconsistencies. The target difference image is selected corresponding to the maximum sum which is further processed to evaluate the MTPM based second-order statistical feature. The recommended approach is assessed on different images from UCID dataset and various social networking websites based on scalar-based and SVM/machine learning based forensic detectors. The experiment results confirm that the projected method offers improved efficiency in comparison to the existing techniques based on different forgery scenarios.

1. Introduction

Digital image forensics is used to acquire the key information related to digital images history which includes acquisition, coding and editing phase. Mining of this important historical information helps to discover the legitimacy of images. A digital forgery modifies the fundamental image statistics but visually imperceptible. These statistical modifications can be utilized as traces in the digital investigation for the verification of digital images (Fan et al., 2017; Singh et al., 2019). A few techniques are available related to the JPEG compression detection (Bianchi and Piva, 2011, 2012; Wang et al., 2011) which can provide a proper localization of forgery in digital images. Most of the previous techniques require a large size image portion for the statistical analysis, so proper localization of tampered regions is not possible with small size images.

The digital imaging camera process comprises of two stages i.e. image acquisition phase and processing phase as shown in Fig. 1. In the image acquisition process, the light coming from the outside three-dimensional world is filtered by the color filter array (CFA). Afterwards, the filtered light falls on the charge couple device (CCD) sensor. In this process, each pixel is allotted with one particular color only and rest two colors are acquired by applying the interpolation process for each pixel. This interpolation process introduces demosaicing artifacts and these artifacts become inconsistent due to image tampering. Therefore, the authenticity of an image is evaluated based on the presence or absence of inconsistencies in demosaicing artifacts. Afterwards, the resultant image acquired from the acquisition phase has undergone through pre-processing in which the image is encoded for the purpose of data compression, intensity variations and then stored in the camera memory.

Swaminathan et al. proposed a camera identification technique (Swaminathan et al., 2007) in order to estimate the engaged CFA pattern and interpolation kernel. The inconsistencies exploited by the same authors among the demosaicing parameters put the integrity of an image in doubt (Swaminathan et al., 2008). A correlation model based on the second-order derivative is employed by Cao and Kot in (Cao and Kot, 2009) for the classification of various demosaicing artifacts based approaches. In (Bayram et al., 2008), Bayram et al. analyzed the features proposed in (Popescu and Farid, 2005a) and (Gallagher, 2005) to detect and classify traces of demosaicing for the identification of the source camera

* Corresponding author.
E-mail address: ksingh@thapar.edu (K. Singh).

https://doi.org/10.1016/j.fsidi.2019.200899
1742-2876/

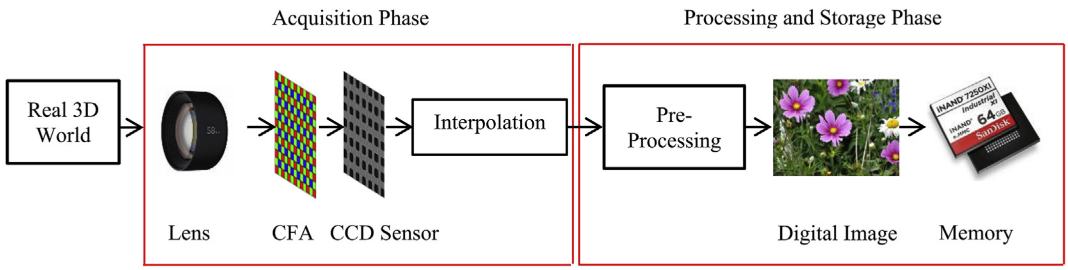


Fig. 1. Digital image formation process.

model. Moreover, a neural network framework is proposed by Fan et al. in (Fan et al., 2009) for image authentication.

For the detection of demosaicing traces, a scheme is recommended by Popescu et al. to identify the interpolation footprints introduced during the processes of resampling (Popescu and Farid, 2005b) and demosaicing (Popescu and Farid, 2005a) on digital images. In this technique, the expectation and maximization process is used to calculate the parameters of the interpolation kernel. In (Gallagher, 2005), for the detection of demosaicing traces, Gallagher found that variance of the second-order derivative performed on the signal processed through interpolation is periodic in nature. Then, the second derivative periodicity on the whole image is examined through the analysis of the Fourier transform. A method is proposed by Gallagher and Chen in (Gallagher and Chen, 2008) based on the Fourier analysis of high pass filtered image in order to capture the presence of periodic footprints in the interpolated coefficients variance.

In (Dirik and Memon, 2009), the sensor noise power of an image i.e. the difference between the interpolated and attained pixels is used by Dirik and Memon to confirm the existence of demosaicing pixels. The features proposed in (Dirik and Memon, 2009) are based on 96×96 blocks, therefore does not provide fine-localization of tampered regions. In (Kirchner, 2008) and (Kirchner and Gloe, 2009), Kirchner exploits that the process of revealing the periodic artifacts is independent of genuine prediction weights of resampling filter. The derivatives of the image which is processed with interpolation operation are considered for forgery detection by Mahdian and Saic in (Mahdian and Saic, 2008). A secure authentication system is proposed for digital images by Uehara et al. in (Uehara et al., 2004). Furthermore, Singh and Aggarwal studied various forgery detection techniques based on videos in (Singh and Aggarwal, 2018). A fine-grained image tampering localization approach is proposed by Ferrera et al. in (Ferrara et al., 2012) by analyzing the CFA artifacts. Moreover, a Markov model based image tampering detection approach is proposed in (Singh et al., 2018) by Singh et al. to analyze the CFA artifacts. An effective forensic approach is presented by Li et al. in (Li et al., 2018) based on the residual domain attributes to recognize the different image operations.

The first-order statistical analysis based forensic techniques can be fooled easily by applying some anti-forensic attacks correspond to the particular image operation. Therefore, a higher-order statistical analysis is required to identify the variations created during the image tampering. Moreover, higher-order statistical analysis in difference domain is limited in the existing forensic techniques. A detection scheme is presented in this paper based on the Markov model in difference domain to discover the tampered regions in images by investigating the discrepancies of demosaicing artifacts.

The following are the major contributions of this paper:

- There is an inevitable modification of original image pixel values during the forgery creation. Thus, the inherent image statistics such as adjacent pixels correlation cannot be preserved. Therefore, in the proposed scheme CFA artifacts inconsistencies analysis is performed in the difference domain due to the less dependence on the image contents as compared to the spatial domain.
- The higher-order statistical analysis based on MTPM further exploits the statistical CFA artifacts inconsistencies in the difference domain by reflecting the variations in neighboring DCT coefficients density.
- The experimental results indicate that the projected forensic technique offers better results than the existing methods in terms of detection accuracy and forged region localization.
- Moreover, the computation cost of the suggested method is less in comparison to the previous forgery detection techniques in terms of processing time.

The rest of the paper is arranged as follows. Section 2 discusses the suggested forgery revealing method and results are defined in Section 3. Lastly, Section 4 concludes the paper.

2. Proposed scheme

During the image acquisition phase, the CFA interpolation of the source camera introduces a strong correlation between the neighboring pixels of an image. This correlation gets damaged during the creation of forgery. Therefore, the proposed approach exploits the CFA interpolation or demosaicing artifacts by performing higher-order statistical analysis based on the Markov transition probability matrix (MTPM) in order to detect the image tampering. In the proposed scheme as revealed in Fig. 2, the higher-order investigation is performed in difference domain in place of the spatial domain due to the less dependence of the difference domain on image contents.

2.1. Selection of target difference image

In the proposed scheme, the given image is initially re-interpolated by considering various CFA patterns. There are 36 different CFA pattern arrangements for a 2×2 CFA cell. Out of these 36 combinations, most of the available digital cameras utilize 4 Bayer CFA arrangements as revealed in Fig. 3. Thus, in the recommended technique, these 4 Bayer CFA patterns i.e. $[2\ 1; 3\ 2], [2\ 3; 1\ 2], [3\ 2; 2\ 1], [1\ 2; 2\ 3]$ are considered for re-interpolation. Note that the bi-cubic interpolation does not change the detection results significantly, so we use bilinear interpolation for simplicity. Now, the difference between the given image and its corresponding re-interpolated versions is estimated in order to analyze the difference domain by using the following equation:

$$I_{diff}\left(m,n,p_{cfa}\right) = \frac{1}{3}\sum_{c=1}^{3}\left[I(m,n,c) - I_{p_{cfa}}(x,y,c)\right]^2 \tag{1}$$

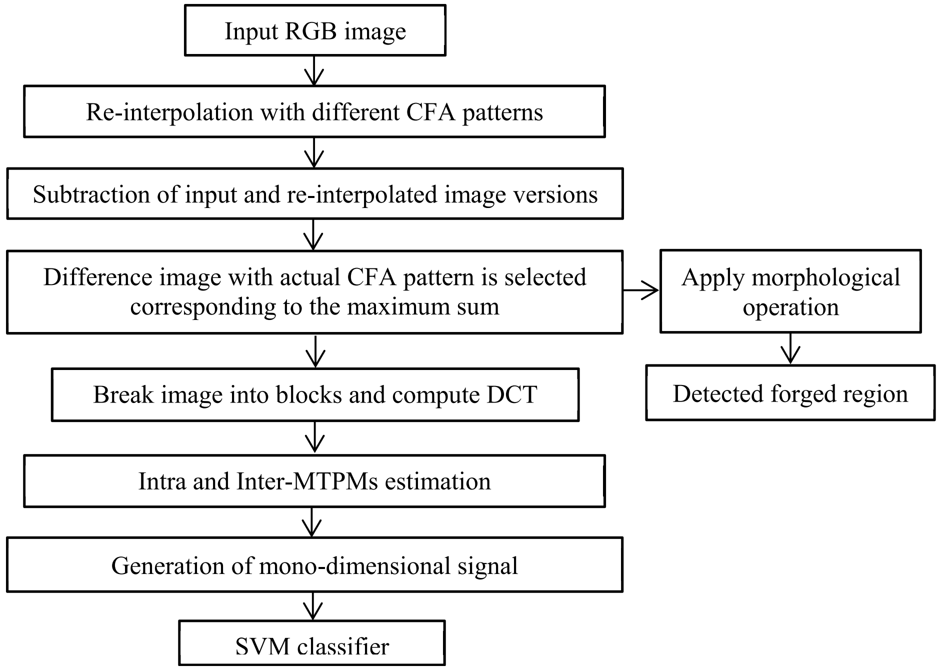


Fig. 2. Sketch of the proposed forgery detection approach based on higher-order statistics.

Fig. 3. Different Bayer CFA patterns used for re-interpolation in the proposed scheme.

where $I(.)$ represents the given three channels RGB color image and $c = 1, 2, 3$ denotes the three channels. The function $I_{p_{cfa}}$ is the re-interpolated version of the original image. The difference images show a significant difference within and outside the forged region due to the underlying image contents. Thus, the difference domain can act as an important measure for forensic investigation. The target difference image is selected corresponding to the maximum sum from the pool of difference images based on different CFA patterns, thereby providing the maximum discrepancy between the forged and the remaining original portion of the considered image. The selected difference image is further processed through morphological operation in order to generate the perfect localization of the forged region.

2.2. Evaluation of Markov transition probability matrices in the DCT domain

The strong intra and inter-block relationship between the coefficients can be exploited by using the MTPM (Lu et al., 2015) statistical feature. The intra-block correlation is the dependence between the DCT coefficients in the same block along horizontal, vertical and diagonal directions. Conversely, the inter-block dependency is the correlation among the coefficients at the same location of the neighboring block along horizontal, vertical and diagonal directions. In the proposed scheme, the concept of MTPM is used to highlight the CFA artifacts variations for forgery detection.

Initially, a 2-D array of 8×8 block absolute DCT coefficients as shown in Fig. 4(a) is obtained from the selected target difference image (I_{tar}) of size $m \times n$ acquired in Section 2.1. So, DCT coefficient 2-D array denoted by $D'(f', e')$ has $(m/8) \times (n/8)$ total number of DCT blocks. Afterward, difference DCT coefficient 2-D arrays along horizontal, vertical, main and minor diagonal directions as shown in Fig. 5 are used to enhance the CFA artifacts inconsistencies exposed by the difference domain. Then, in order to employ the second-order statistics for forgery detection, the Markov method is utilized to formulate the difference 2-D arrays. The intra-block MTPMs for the horizontal and vertical adjacent DCT coefficients are computed by Eqs. (2) and (3). $\boldsymbol{MTPM}_{intra,h}$ and $\boldsymbol{MTPM}_{intra,v}$ denote the Markov transition matrices for horizontal and vertical directions having conditional distribution probabilities $\mathcal{P}(D'(f'+1, e') = v | D'(f', e') = u)$ and $\mathcal{P}(D'(f', e'+1) = v | D'(f', e') = u)$, respectively for the coefficients pair (v, u). The value of the function $\partial(a, b) = 1$ if and only if $a = b$; otherwise, $\partial(a, b) = 0$. Similarly, the

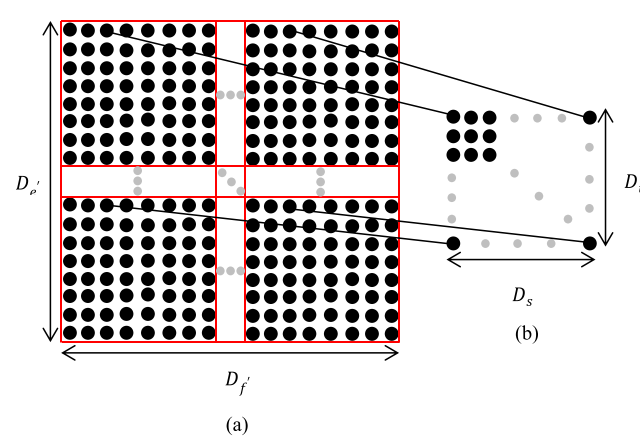


Fig. 4. (a) DCT coefficient 2-D array, (b) Formation of Mode (3) 2-D array.

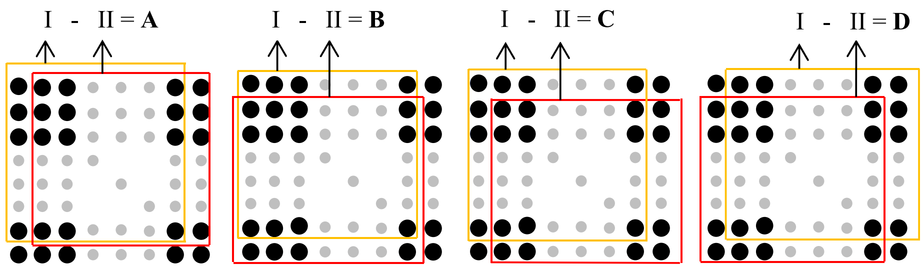


Fig. 5. Horizontal difference DCT/Mode 2-D array (**A**), Vertical difference DCT/Mode 2-D array (**B**), Main diagonal difference DCT/Mode 2-D array (**C**), and Minor diagonal difference DCT/Mode 2-D array (**D**).

conditional distribution probabilities for intra-block are formulated for main and minor diagonal difference arrays.

$$\boldsymbol{MTPM}_{intra,h}(u,v)=\mathcal{P}\left(D'\left(f'+1,e'\right)=v\middle|D'\left(f',e'\right)=u\right)=\frac{\mathcal{P}\left(D'\left(f',e'\right)=u,D'\left(f'+1,e'\right)=v\right)}{\mathcal{P}\left(D'\left(f',e'\right)=u\right)}=\frac{\sum_{e'=1}^{D_{e'}-1}\sum_{f'=1}^{D_{f'}-1}\partial\left(D'\left(f',e'\right),u\right)\partial\left(D'\left(f'+1,e'\right),v\right)}{\sum_{e'=1}^{D_{e'}-1}\sum_{f'=1}^{D_{f'}-1}\partial\left(D'\left(f',e'\right),u\right)} \tag{2}$$

$$\boldsymbol{MTPM}_{intra,v}(u,v)=\mathcal{P}\left(D'\left(f',e'+1\right)=v\middle|D'\left(f',e'\right)=u\right)=\frac{\mathcal{P}\left(D'\left(f',e'\right)=u,D'\left(f',e'+1\right)=v\right)}{\mathcal{P}\left(D'\left(f',e'\right)=u\right)}=\frac{\sum_{e'=1}^{D_{e'}-1}\sum_{f'=1}^{D_{f'}-1}\partial\left(D'\left(f',e'\right),u\right)\partial\left(D'\left(f',e'+1\right),v\right)}{\sum_{e'=1}^{D_{e'}-1}\sum_{f'=1}^{D_{f'}-1}\partial\left(D'\left(f',e'\right),u\right)} \tag{3}$$

The inter-block relationship is revealed among the DCT coefficients modes i.e. the coefficients situated in equivalent location in the 8 × 8 blocks, which capture the frequency characteristics of those blocks. Initially, the mode 2-D arrays are created from the DCT coefficient 2-D array. Subsequently, difference mode 2-D arrays along different directions are generated from the mode 2-D arrays of the considered image in order to reveal the inter-block correlation as exposed in Fig. 5. These difference mode 2-D arrays are demonstrated by the Markov method. Afterward, the average transition probability matrix is evaluated for all difference mode 2-D arrays along each direction. The alignment of each mode from all the 8 × 8 blocks provides the mode 2-D arrays. The creation of mode (3) 2-D can be understood from Fig. 4(b). The size of each mode 2-D array is $D_s = D_{f'}/8$ in horizontal direction and $D_t = D_{e'}/8$ in the vertical direction. For the particular image, 63 mode 2-D arrays are obtained but mode 1 i.e. DC mode is excluded because the results are slightly affected by the inclusion of mode 1. The mode coefficient in a mode 2-D array is represented as $D^{(mode)}(s,t)$, where $s \in [0, D_s - 2]$, $t \in [0, D_t - 2]$ and $mode \in [2, 64]$. The difference mode 2-D array along horizontal, vertical, main and minor diagonal directions are calculated as:

$$D_h^{(mode)}(s,t) = D^{(mode)}(s,t) - D^{(mode)}(s+1,t) \tag{4}$$

$$D_v^{(mode)}(s,t) = D^{(mode)}(s,t) - D^{(mode)}(s,t+1) \tag{5}$$

$$D_{d'}^{(mode)}(s,t) = D^{(mode)}(s,t) - D^{(mode)}(s+1,t+1) \tag{6}$$

$$D_{d''}^{(mode)}(s,t) = D^{(mode)}(s+1,t) - D^{(mode)}(s,t+1) \tag{7}$$

The inter-block MTPM along horizontal and vertical directions is calculated by applying the Eqs. (8) and (9). $\boldsymbol{MTPM}_{inter,h}$ and $\boldsymbol{MTPM}_{inter,v}$ denote the MTPM features components with conditional distribution probabilities $\mathcal{P}(D_h(s+1, t) = v|D_h(s, t) = u)$ and $\mathcal{P}(D_v(s, t+1) = v|D_v(s, t) = u)$, respectively. Likewise, the inter-block conditional distribution probabilities are formulated for main and minor diagonal difference arrays in the DCT domain. Each MTPM contains $(H-G+1)^2$ number of elements, where $[G, H]$ is the range of u and v. The matrix comprising $(H-G+1)^2$ number of feature components is evaluated by using the Eq. (10). The effect of DCT coefficients modification caused by the CFA artifacts variation dominates nearby the origin. So, the statistical range of u and v is adjusted to $[-4, 4]$ which results in the 81 feature components for each MTPM. Thus, we get 648 feature components in total from all the MTPMs.

$$\begin{aligned}\boldsymbol{MTPM}_{inter,h}(u,v) &= \mathcal{P}(D_h(s+1,t)=v|D_h(s,t)=u) = \frac{\mathcal{P}(D_h(s,t)=u,\ D_h(s+1,t)=v)}{\mathcal{P}(D_h(s,t)=u)} \\ &= \frac{\sum_{t=1}^{D_t-2}\sum_{s=0}^{D_s-2}\sum_{mode=2}^{64}\partial\left(D_h^{(mode)}(s,t)=u\right)\partial\left(D_h^{(mode)}(s+1,t)=v\right)}{\sum_{t=1}^{D_t-2}\sum_{s=0}^{D_s-2}\sum_{mode=2}^{64}\partial\left(D_h^{(mode)}(s,t)=u\right)}\end{aligned} \tag{8}$$

$$\boldsymbol{MTPM}_{inter,v}(u,v) = \mathcal{P}(D_v(s,t+1)=v|D_v(s,t)=u) = \frac{\mathcal{P}(D_v(s,t)=u,\ D_v(s,t+1)=v)}{\mathcal{P}(D_v(s,t)=u)}$$
$$= \frac{\sum_{t=1}^{D_t-2}\sum_{s=0}^{D_s-2}\sum_{mode=2}^{64}\partial\left(D_v^{(mode)}(s,t)=u\right)\partial\left(D_v^{(mode)}(s,t+1)=v\right)}{\sum_{t=1}^{D_t-2}\sum_{s=0}^{D_s-2}\sum_{mode=2}^{64}\partial\left(D_v^{(mode)}(s,t)=u\right)} \tag{9}$$

$$\boldsymbol{MTPM} = \begin{bmatrix} \mathcal{P}(G|G) & \mathcal{P}(G|G+1) & \cdots & \mathcal{P}(G|H) \\ \mathcal{P}(G+1|G) & \mathcal{P}(G+1|G+1) & \cdots & \mathcal{P}(G+1|H) \\ \vdots & \vdots & \ddots & \vdots \\ \mathcal{P}(H|G) & \mathcal{P}(H|G+1) & \ldots & \mathcal{P}(H|H) \end{bmatrix} \tag{10}$$

The features components are converted into a mono-dimensional signal of dimensionality (1, 648) by concatenating the rows of all intra and inter-block MTPMs revealing the existence of CFA artifacts discrepancies introduced at the time of forgery creation. This signal is used as a resultant second-order statistical feature vector for the detection of forged images based on the CFA artifacts. This feature matrix is given as SVM classifier input for training on a given set of images. Another set of completely different images is constructed for testing purpose and feature matrix is extracted from them. SVM classifier separates the forged and original images by making a decision boundary that separates them as far as possible.

Algorithm 1. Algorithm for the proposed forgery detection scheme based on CFA artifacts:

Input:
I: Input RGB image under investigation
Output:
L_{map}: Forged region localization
H_{final}: Resultant feature vector based on MTPMs
Parameters:
p_{cfa}: 4 Bayer CFA patterns
$I_{p_{cfa}}$: Re-interpolated image
I_{diff}: Difference image
I_{tar}: Target difference image
begin
$[m,n] = size(I);$
$p_{cfa} = \{[2\ 1; 3\ 2]\ [2\ 3; 1\ 2]\ [3\ 2; 2\ 1]\ [1\ 2; 2\ 3]\};$
$Max_{sum} = zeros(1, p_{cfa});$
$I_{diff} = zeros(m, n, length(p_{cfa}));$
for $i = 1{:}\ length(p_{cfa})$ ***do***
$BinFilter = [];$
$I_{diff} = zeros(m,n,4);$
$CFA = p_{cfa}\{i\};$
$R = CFA == 1;$
$G = CFA == 2;$
$B = CFA == 3;$
$BinFilter(:,:,1) = repmat(R, size(I,1)/2, size(I,2)/2);$
$BinFilter(:,:,2) = repmat(G, size(I,1)/2, size(I,2)/2);$
$BinFilter(:,:,3) = repmat(B, size(I,1)/2, size(I,2)/2);$
$CFA_I = double(I).*\ BinFilter;$
$I_{p_{cfa}} = bilinInterp(CFA_I, BinFilter, CFA);$
for $c = 1:3$ ***do***
$I_{diff}(:,:,i) = I_{diff}(:,:,i) + (I(:,:,c) - double(I_{p_{cfa}}(:,:,c))).\hat{}2;$
end
$I_{diff}(:,:,i) = I_{diff}(:,:,i)\ /\ 3;$
$Max_{sum}(1,i) = sum(sum(I_{diff}(:,:,1)));$
end
$[value, index] = max(Max_{sum});$
$I_{tar} = I_{diff}(:,:,index);$
$L_{map} = imfill(I_{tar});$
$I_{dct} = abs(dct(I_{tar}));$
$f1 = MTPM_{intra,h}(I_{dct});\ f2 = MTPM_{inter,h}(I_{dct});\ f3 = MTPM_{intra,v}(I_{dct});\ f4 = MTPM_{inter,v}(I_{dct});$
$f5 = MTPM_{intra,d'}(I_{dct}); f6 = MTPM_{inter,d'}(I_{dct});\ f7 = MTPM_{intra,d''}(I_{dct});\ f8 = MTPM_{inter,d''}(I_{dct});$
$H_{final} = [f1;\ f2;\ f3;\ f4;\ f5;\ f6; f7; f8];$
end

3. Experiment results

The evaluation of the suggested scheme is performed by considering the UCID dataset (Schaefer and Stich, 2003) of 1338 uncompressed original color images of TIFF format. This dataset was created by taking images from Minolta Dimage 5 digital color camera. These images contain demosaicing footprints due to the employment of Bayer Color Filter Array in the considered camera with the unknown in-camera demosaicing algorithm. Moreover, the proposed scheme is further evaluated on the doctored images viral on various social websites such as Facebook, Snapchat, Twitter, etc. The proposed approach is capable to differentiate between the original and fake images. During the creation of image forgery, counterfeiter applies different image processing operations such as spatial filtering, edge smoothing, scaling, etc. in order to make the forged image identical to the original one. These image operations significantly modify the CFA artifacts of the tampered portion; result in the easier detection of the forged region.

The effectiveness of the presented forensic approach is estimated by performing a comparative analysis with existing techniques such as Gallagher and Chen (GC-B and GC-L) (Gallagher and Chen, 2008), Dirik and Memon (DM) (Dirik and Memon, 2009), Ferrara et al. (FBRP) (Ferrara et al., 2012), Singh et al. (SSS) (Singh et al., 2018), and Li et al. (LLQH) (Li et al., 2018). For testing, images with bilinear interpolation are considered and morphological operations are applied on the selected difference image for the localization of forged areas in the image under investigation. The proposed algorithm offers improved results than the existing techniques as displayed in Fig. 6. This happens because a higher-order statistical investigation is performed based on MTPM to identify the inconsistencies in CFA footprints. The detection process is significantly disturbed due to the application of JPEG compression is one of the main issues in the existing forensic techniques. The detection performance declines at higher compression or decompression rates. Thus, the evaluation of the projected method is further extended to JPEG compression scenarios.

In steganalysis approaches (Holub and Fridrich, 2013; Pevny et al., 2010), the efficiency of various forensic detectors is validated against different forgeries by using minimum decision error (P_e). Primarily, the receiver operating characteristic (ROC) curve is plotted by considering different SVM-based forensic detectors. Two different classes i.e. positive and negative are considered by the SVM classifier for classification purpose. The positive class indicates the forged images and negative class signifies the

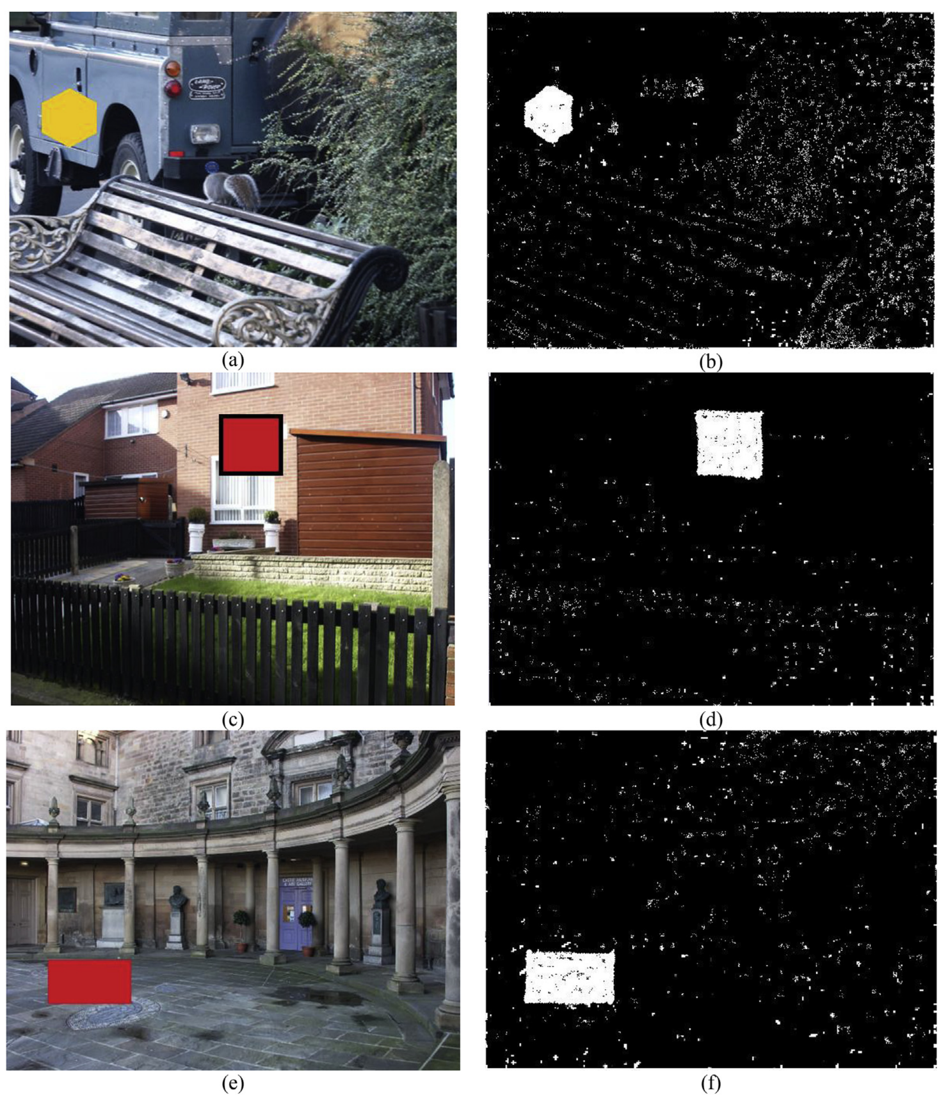


Fig. 6. (a), (c), and (e) Forged images created from UCID dataset; (b), (d), and (f) Tampered region detection using proposed forensic technique.

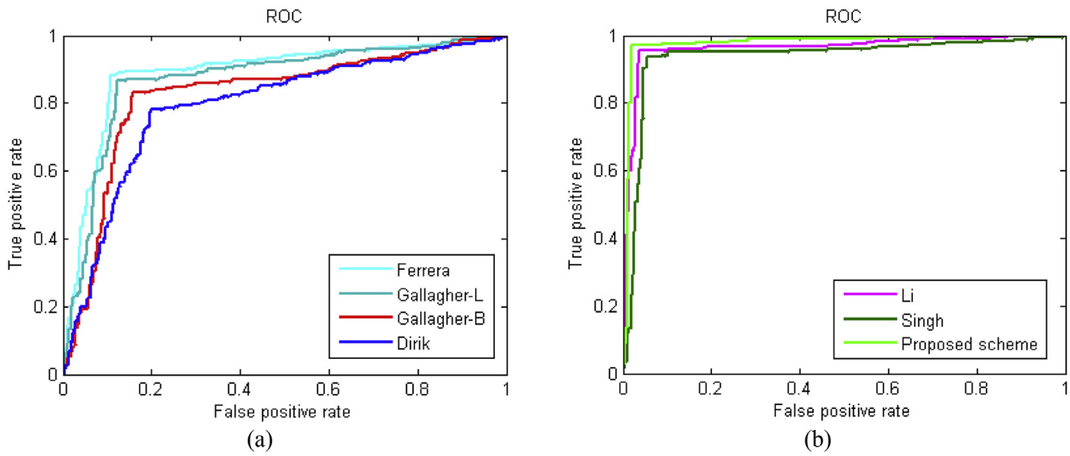


Fig. 7. ROC curves for (a) Scalar-based forensic detectors based on CFA artifacts, (b) SVM-based forensic detectors, when tested against various types of forgeries. It is worth noting that forensic detectability rises as the ROC curve approach to the upper left corner.

original images. Subsequently, the minimum decision error (P_e) is calculated from the ROC curve. This error is denoted by a point on ROC curve with least number of inaccurately categorized images.

In this section, we are considering the forged images, which do not suffer from post-processing attacks. For forgery creation, an arbitrary part of any image is taken and is pasted over the same

Table 1
Comparison between the presented forensic method and existing methods based on average minimum decision error values by considering different forgeries.

	Gallagher (GC-B) (Gallagher and Chen, 2008)	Gallagher (GC-L) (Gallagher and Chen, 2008)	Dirik (Dirik and Memon, 2009)	Ferrara (Ferrara et al., 2012)	Singh (Singh et al., 2018)	Li (Li et al., 2018)	Proposed scheme
Minimum decision error (P_e)	0.3247	0.3126	0.3485	0.2718	0.1429	0.1036	0.0751

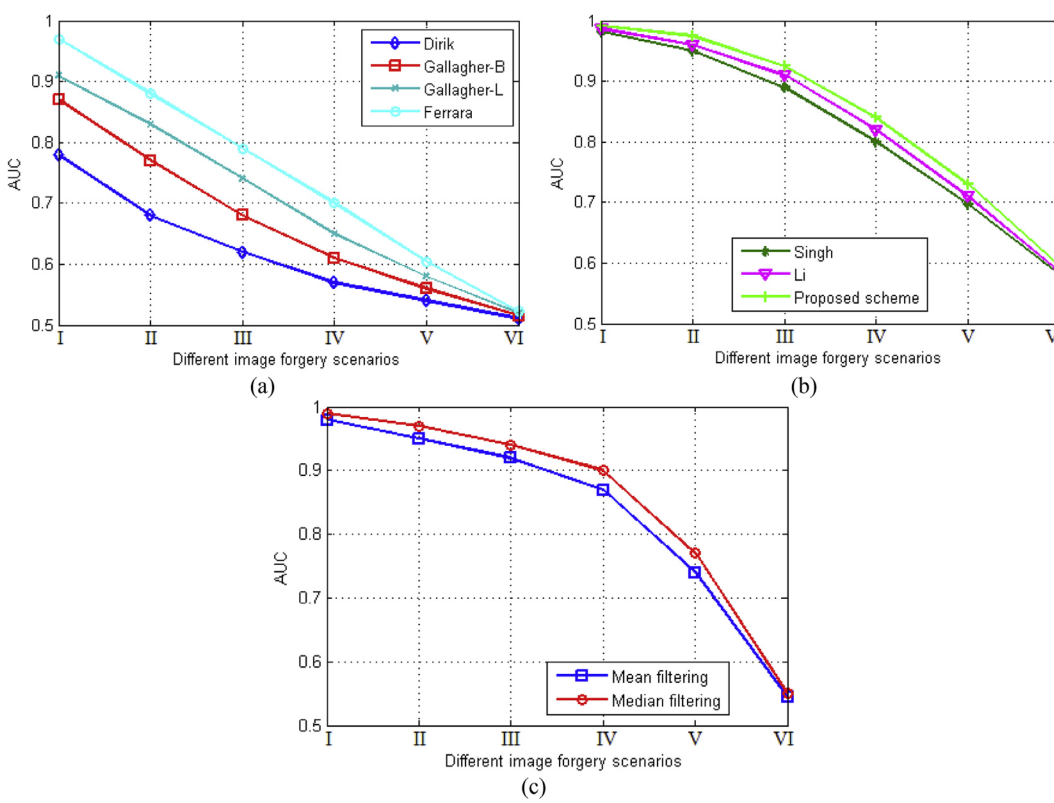


Fig. 8. The comparative analysis based on AUC values under different forgeries scenarios i.e. (I) Bilinear interpolation, (II) In-camera demosaicing and the remaining cases are under JPEG compression with compression quality of (III) 100%, (IV) 95%, (V) 90% and (VI) 85%. (a) Scalar-based detectors, (b) SVM-based detectors and (c) When images from different scenarios are processed with Mean and Median filtering.

image in an unpredictable manner. In order to calculate the accuracy, we have given the complete dataset of 1338 images to the classifier by dividing half number of images into a training set and another half into the testing set.

The forensic training/testing is performed by considering both the tampered and original images. The training of the SVM classifier is done by using the LIBSVM (Chang and Lin, 2011) with a Gaussian kernel. The factors of SVM classifier are attained by using five-fold cross-validation with the multiplicative grid (Pevný et al., 2010). The SVM classifiers are trained by using the original as well as their corresponding forged images. Subsequently, forensic testing is performed by using these classifiers based on the images generated from the considered dataset.

The presented framework aims to detect the image forgeries by revealing the CFA artifacts irregularities. Forged and original images are two different cases considered in SVM classifier. The number of tampered images that are properly categorized under the doctored image class is denoted by the true positive rate (TPR). Similarly, the number of tampered images that are wrongly categorized under original image class is described by the false positive rate (FPR). Most of the existing forensic techniques analyzing the CFA artifacts for tampering detection are based on a single scalar feature. Therefore, ROC curves for the scalar-based detectors and SVM-based machine learning detectors are provided separately as shown in Fig. 7 in order to make the comparison feasible. The ROC curve corresponding to the presented forensic algorithm is closer to the upper left corner as compared to the considered (scalar or SVM-based) existing techniques as presented in Fig. 7. Thus, the proposed scheme offers enhanced forensic detectability when compared to existing methods, when a test is conducted on various types of tampered images. The presented forensic approach provides smaller minimum decision error values in comparison to existing approaches, so able to detect the various forgeries proficiently as shown in Table 1.

Note that most of the existing CFA artifacts based forensic approaches are not able to perfectly localize the forged region in the case of forged images processed with JPEG compression. Therefore, the recommended approach is also evaluated by considering the forged images processed with JPEG compression operation, Mean and Median filtering operations. Fig. 8 provides the comparative analysis of the proposed scheme under different image forgery scenarios with both the scalar and machine learning based forensic detectors. The first scenario (I) is based on the original images with bilinear interpolation, second (II) on the in-camera demosaicing and remaining scenarios are based on the JPEG compression with compression rates i.e. (III) 100%, (IV) 95%, (V) 90%, and (VI) 85%. It is perceived from Fig. 8 that the proposed second-order statistical feature outperforms the existing techniques in terms of area under

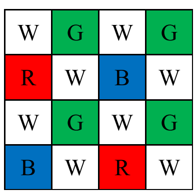
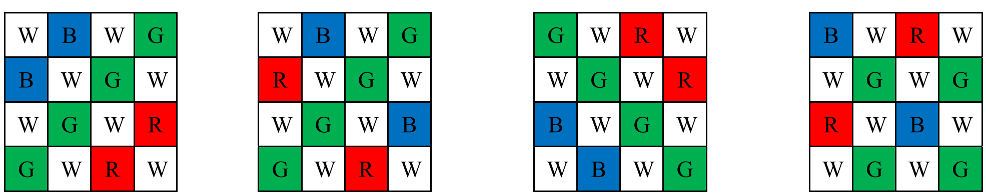


Fig. 9. Sony-RGBW CFA pattern.



Kodak-RGBW CFA pattern   Sony 2012-RGBW CFA pattern   Omnivision-RGBW CFA pattern   Sony 2014-RGBW CFA pattern

Fig. 10. Different RGBW CFA patterns used for re-interpolation in the proposed scheme.

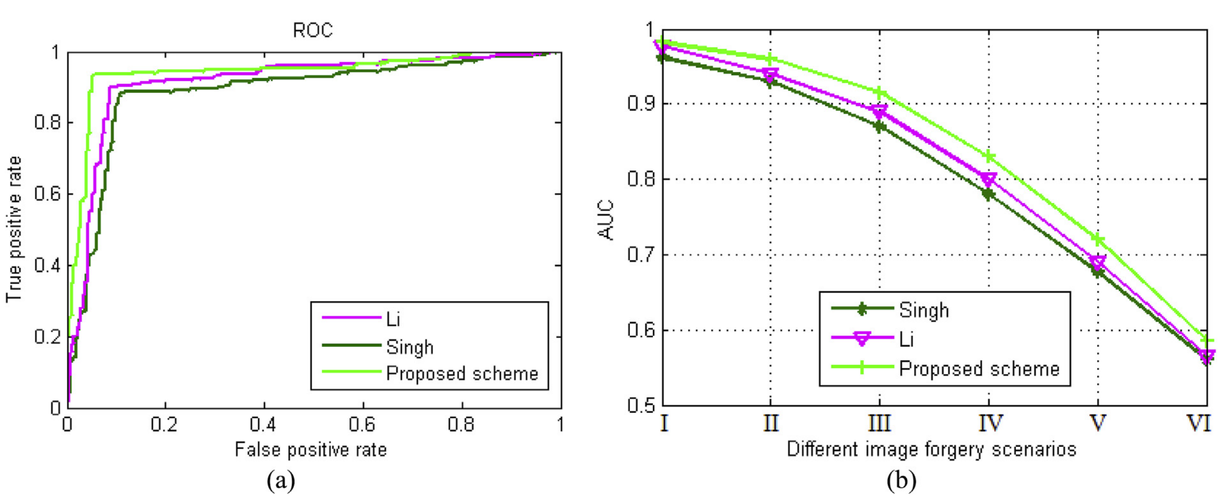


Fig. 11. (a) ROC curves for various SVM-based forensic detectors, when tested against various types of forgeries. (b) Comparative analysis of SVM-based forensic detectors based on AUC values under different forgeries scenarios i.e. (I) Bilinear interpolation, (II) In-camera demosaicing and the remaining cases are under JPEG compression with compression quality of (III) 100%, (IV) 95%, (V) 90% and (VI) 85%.

the curve (AUC) values, when tested under different forgery scenarios.

3.1. Evaluation of proposed scheme on RGBW CFA pattern

There are many non-standard CFA patterns available in the literature such as RGBE (Bayer-like with one of the green filters changed to emerald), RYYB (One red, two yellow, and one blue), CYYM (One cyan, two yellow, and one magenta), CYGM (One cyan, one yellow, one green, and one magenta), and RGBW (Bayer-like with one of the green filters changed to white). It is worth noting that most of these CFA patterns are not commercially used. However, the proposed forgery detection approach is further evaluated by considering the RGBW CFA pattern (for example, Huawei P8 mobile camera) on Kodak dataset (Dataset Eastman Kodak [Online], 2014). We have reconstructed the Kodak images by using the Sony-RGBW CFA pattern as provided in Fig. 9 for evaluation purposes. In the proposed methodology, the re-interpolation (bilinear) is performed with four different forms of RGBW CFA pattern as shown in Fig. 10. Although, RGBW has some robustness against noise and low light conditions, it is not popular and does not have good performance as compared to the standard Bayer pattern (Kwan and Chou, 2019).

It can be observed from Fig. 11(a) that ROC curve for the proposed scheme is closer to the upper left corner in comparison to the other existing techniques, when evaluation is performed on the RGBW CFA pattern images. Therefore, the proposed forensic scheme based on second-order statistical analysis provides better detection accuracy as compared to the existing techniques. Moreover, the proposed technique outperforms the existing methods in

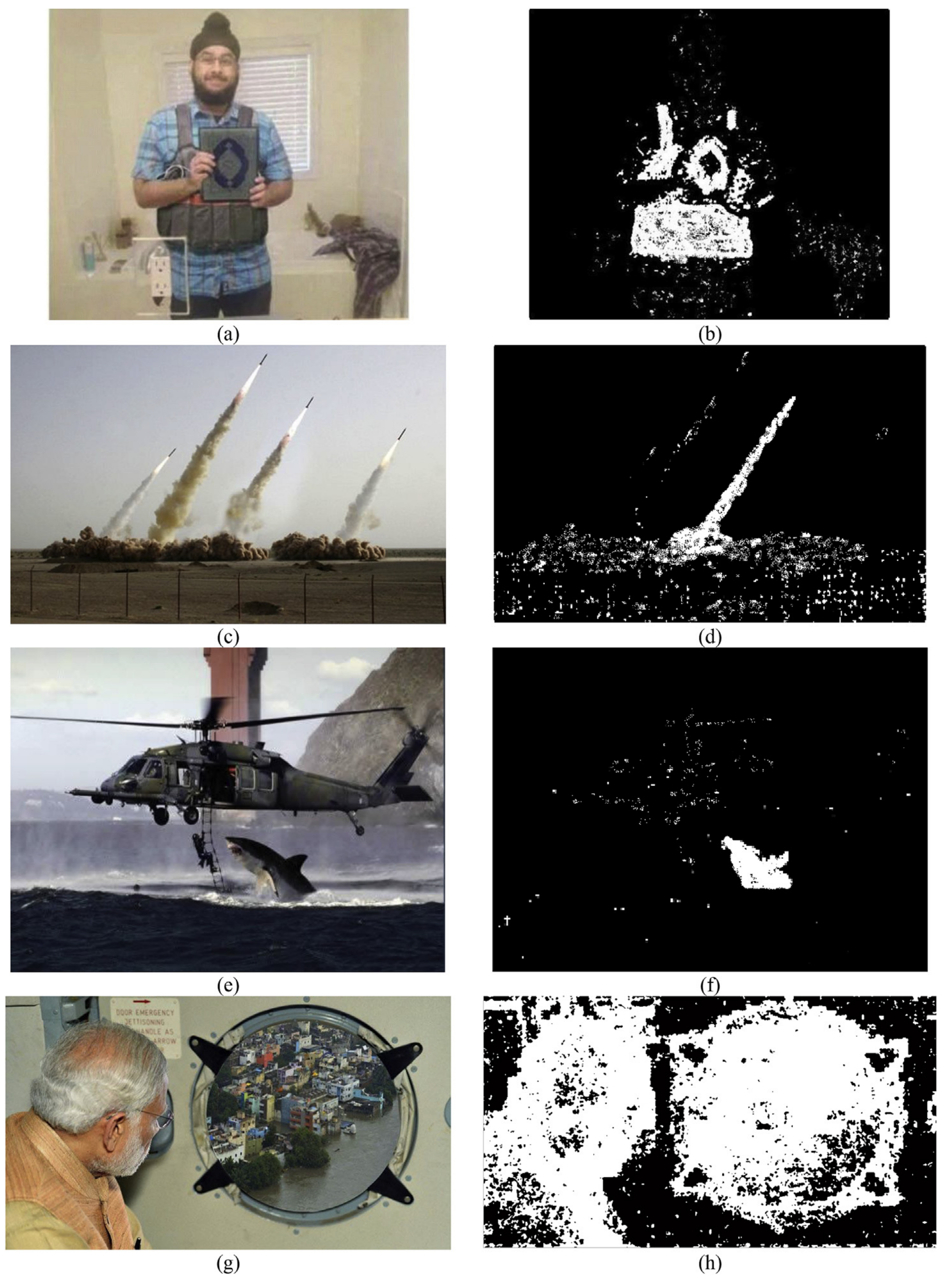


Fig. 12. Forged images from different social networking sites (a) Sikh boy (https://mashable.com, 2016), (c) Missile (https://www.pocket-lint.com, 2008), (e) Shark (https://www.pocket-lint.com, 2001), and (g) PM Narendra Modi (https://indianexpress.com, 2015); (b), (d), (f), and (h) Forgery region detection by proposed forensic technique.

Table 2
Comparison of the suggested approach and existing schemes based on the average computation time (sec) by considering various image forgeries.

Different forensic approaches	Average detection time (sec)
Gallagher (GC-B) (Gallagher and Chen, 2008)	202.75
Gallagher (GC-L) (Gallagher and Chen, 2008)	204.64
Dirik (Dirik and Memon, 2009)	206.18
Ferrara (Ferrara et al., 2012)	194.04
Singh (Singh et al., 2018)	168.32
Li (Li et al., 2018)	161.45
Proposed scheme	157.27

terms of AUC values, when tested under different forgery scenarios as shown in Fig. 11(b).

3.2. Evaluation of proposed scheme on realistic images

The objective of forensic algorithms is to restrict the fake images from uploading to social networking sites. Therefore, in this paper, we have validated our proposed image tampering detection scheme by conducting a test on the realistic images from the various social networking sites as exposed in Fig. 12. Hence, this evaluation of proposed forgery detection scheme on realistic images has a great relevance to the forensic community. The proposed method is capable in the detection of tampered regions as revealed in Fig. 12. But, the performance of the suggested scheme decreases with the reduction of tampered region size. The proposed forgery detection scheme is evaluated on MATLAB R2016a by using a PC with 2.13 GHz CPU and 3 GB RAM. With the growth in the size of an image, the forgery detection time also increases. The average detection time of the recommended technique is smaller than the existing approaches as exposed in Table 2.

3.3. Limitation analysis

Now, we summarize and examine the limitations of proposed forensic technique on the basis of experimental results and discussions.

- The efficacy of the proposed technique drops with the decrease in the size of forged region due to the inadequate image statistics. This may have adverse effects on some forensic applications such as tampered region localization.
- Automatic detection may give a significant false positive rate due to the presence of uniform regions, when test is conducted on realistically forged images.
- The suggested technique may not work well, if the adversary applies re-demosaicing after tampering.
- The images taken with X3 Foveon sensors do not exhibit any CFA demosaicing artifacts. Therefore, the proposed method does not work for images attained with X3 Foveon sensors.
- The recommended image tampering detection scheme may not be directly applicable to cameras with super CCD.
- The proposed technique is sensitive to JPEG compression and resizing operation because these types of operations distort or suppress CFA artifacts. Thus, CFA based forgery detection may not be successful after these operations.

4. Conclusions

This paper presents the image forgery detection technique based on MTPM by analyzing CFA artifacts discrepancies without any prior knowledge of the location of forged region. The second-order statistical analysis based on MTPM improves the detection accuracy of the forgery map, thereby efficiently detect whether the considered image is tampered or not. The experiment results demonstrate that the presented forensic scheme is superior to the existing methods in terms of detection accuracy and localization of the tampered region along with the reduction in computation time. Moreover, the effectiveness of the recommended approach is also validated by considering the forged images from social media and the images which are processed through different image operations such as JPEG compression, Mean and Median filtering. The adequacy of the suggested method decreases with the reduction of the forged region size. Therefore, in order to resolve this problem, further work can be devoted to employ the convolutional neural networks (CNNs) for the detection of tampered region of small dimensionality.

Declaration of competing interest

- All authors have participated in (a) conception and design, or analysis and interpretation of the data; (b) drafting the article or revising it critically for important intellectual content; and (c) approval of the final version.
- This manuscript has not been submitted to, nor is under review at, another journal or other publishing venue.
- The authors have no conflict of interest with any of the suggested reviewers or organization.

Acknowledgements

This work was supported by Visvesvaraya PHD scheme for Electronics and IT, Ministry of Electronics and Information Technology, Government of India (grant PhD-MLA/4(33)/2015-16/01).

References

Bayram, S., Sencar, H.T., Memon, N., 2008. Classification of digital camera-models based on demosaicing artifacts. Digit. Invest. 5, 46–59.
Bianchi, T., Piva, A., Nov./Dec. 2011. Analysis of non-aligned double JPEG artifacts for the localization of image forgeries. In: Proc. WIFS 2011. Foz do Iguaçu, Brazil.
Bianchi, T., Piva, A., Jun. 2012. Image forgery localization via block-grained analysis of JPEG artifacts. IEEE Trans. Inf. Forensics Secur. 7 (3), 1003–1017.
Cao, H., Kot, A., Dec. 2009. Accurate detection of demosaicing regularity for digital image forensics. IEEE Trans. Inf. Forensics Secur. 4 (4), 899–910.
Chang, C.C., Lin, C.J., 2011. LIBSVM: a library for support vector machines. ACM Trans. Intell. Syst. Technol. 2 (3). Art. ID 27.
Dataset Eastman Kodak, 2014 [Online] Available. http://r0k.us/graphics/kodak/.
Dirik, A.E., Memon, N., 2009. Image tamper detection based on demosaicing artifacts. In: Proc. 16th IEEE Int. Conf. On Image Processing (ICIP '09), pp. 1497–1500.
Fan, N., Jin, C., Huang, Y., 2009. A pixel-based digital photo authentication framework via demosaicking inter-pixel correlation. In: Proc. 11th ACM Multimedia and Security Workshop. MM&Sec '09), pp. 125–129.
Fan, J., Chen, T., Kot, A.C., July 2017. EXIF-white balance recognition for image forensic analysis. Multidimens. Syst. Signal Process. 28 (3), 795–815.
Ferrara, P., Bianchi, T., De Rosa, A., Piva, A., 2012. Image forgery localization via fine-grained analysis of CFA artifacts. IEEE Trans. Inf. Forensics Secur. 7 (5), 1566–1577.
Gallagher, A.C., 2005. Detection of linear and cubic interpolation in JPEG compressed images, 0. In: Proc. Canadian Conf. Computer and Robot Vision, pp. 65–72.
Gallagher, A.C., Chen, T., 2008. Image authentication by detecting traces of demosaicing. In: Proc. IEEE Computer Vision and Pattern Recognition Workshops. CVPRW 2008), pp. 1–8.
Holub, V., Fridrich, J., 2013. Digital image steganography using universal distortion. In: Proc. 1st ACM Workshop Inf. Hiding Multimedia Secur., pp. 59–68.
https://indianexpress.com/article/india/india-news-india/pib-removes-pm-modis-chennai-photo-from-website-after-proven-fake/.
https://mashable.com/2016/07/15/sikh-man-accused-nice-terror-attack-veerender-jubbal/#9hhwbmxkEiqh.
https://www.pocket-lint.com/apps/news/adobe/140252-30-famous-photoshopped-and-doctored-images from-across-the-ages.
Kirchner, M., 2008. Fast and reliable resampling detection by spectral analysis of fixed linear prediction residue. In: Proc. 10th ACM Multimedia and Security Workshop. MM&Sec '08), pp. 11–20.

Kirchner, M., Gloe, T., 2009. On resampling detection in re-compressed images. In: Proc. First IEEE Int. Workshop on Information Forensics and Security, pp. 21–25. Dec. 2009.
Kwan, C., Chou, B., Aug. 2019. Further improvement of debayering performance of RGBW color filter arrays using deep learning and pansharpening techniques. Journal of Imaging 5 (8), 1–14.
Li, H., Luo, W., Qiu, X., Huang, J., 2018. Identification of various image operations using residual-based features. IEEE Trans. Circuits Syst. Video Technol. 28 (1), 31–45.
Lu, J., Liu, F., Luo, X., 2015. A study on JPEG steganalytic features: Co-occurrence matrix vs. Markov transition probability matrix. Digit. Invest. 12, 1–14.
Mahdian, B., Saic, S., Sep. 2008. Blind authentication using periodic properties of interpolation. IEEE Trans. Inf. Forensics Secur. 3 (3), 529–538.
Pevny, T., Filler, T., Bas, P., 2010. Using high-dimensional image models to perform highly undetectable steganography. In: Proc. Of International Workshop Inf. Hiding, pp. 161–177.
Pevný, T., Bas, P., Fridrich, J., Jun. 2010. Steganalysis by subtractive pixel adjacency matrix. IEEE Trans. Inf. Forensics Secur. 5 (2), 215–224.
Popescu, A.C., Farid, H., Oct. 2005. Exposing digital forgeries in color filter array interpolated images. IEEE Trans. Signal Process. 53 (10), 3948–3959 pt. 2.
Popescu, A.C., Farid, H., Feb. 2005. Exposing digital forgeries by detecting traces of resampling. IEEE Trans. Signal Process. 53 (2), 758–767 pt. 2.
Schaefer, G., Stich, M., Dec. 2003. UCID: an uncompressed color image database. In: Proc. SPIE, vol. 5307, pp. 472–480.
Singh, R.D., Aggarwal, N., Mar. 2018. Video content authentication techniques: a comprehensive survey. Multimed. Syst. 24 (2), 211–240.
Singh, A., Singh, G., Singh, K., May 2018. A Markov based image forgery detection approach by analyzing CFA artifacts. Multimed. Tools Appl. 77 (9), 1–20.
Singh, K., Kansal, A., Singh, G., Feb. 2019. An improved median filtering anti-forensics with better image quality and forensic undetectability. “. ” *Multidim. Syst. Sign. Process.*
Swaminathan, A., Wu, M., Liu, K.R., Mar. 2007. Nonintrusive component forensics of visual sensors using output images. IEEE Trans. Inf. Forensics Secur. 2 (1), 91–106.
Swaminathan, A., Wu, M., Liu, K.J.R., Mar. 2008. Digital image forensics via intrinsic fingerprints. IEEE Trans. Inf. Forensics Secur. 3 (1), 101–117.
Uehara, T., Naini, R.S., Ogunbona, P., Mar. 2004. A secure and flexible authentication system for digital images. Multimed. Syst. 9 (5), 441–456.
Wang, W., Dong, J., Tan, T., Nov./Dec. 2011. Exploring DCT coefficient quantization effect for image tampering localization. In: Proc. WIFS 2011. Foz do Iguaçu, Brazil.